

Modern Backend Development Course (Firebase + MongoDB + PHP + MySQL)

Course Focus: Practical backend development focused on Firebase, MongoDB, PHP, MySQL, APIs, Authentication, CRUD Operations, and Deployment.

Course Type:

- Practical Based
 - Industry Oriented
 - Beginner to Intermediate
 - Less Theory, More Practical
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Chapter 1 — Introduction to Backend Development

- What is Backend?
- Frontend vs Backend
- Client → Server → Database Flow
- What is API?
- JSON Basics
- SQL vs NoSQL

Practical: Create simple JSON response and backend setup.

Chapter 2 — Firebase Backend Fundamentals

- Firebase Introduction
- Realtime Database
- Firestore Database
- Firebase Project Structure
- Cloud Database Basics

Practical: Create Firebase project and perform CRUD operations.

Chapter 3 — Firebase Authentication

- User Authentication

- Signup/Login System
- Google Authentication
- Protected Routes
- Backend Security Basics

Practical: Build secure login and signup system.

Chapter 4 — MongoDB & NoSQL

- MongoDB Basics
- Collections & Documents
- MongoDB Atlas
- Connection Strings
- NoSQL Database Concepts

Practical: Create MongoDB cluster and perform CRUD operations.

Chapter 5 — REST APIs & CRUD Operations

- REST API Basics
- GET, POST, PUT, DELETE
- Status Codes
- CRUD Operations
- API Endpoints

Practical: Create Student CRUD API.

Chapter 6 — API Connections & JSON Handling

- Dummy APIs
- JSON Data Handling
- API Requests & Responses
- Frontend and Backend Communication
- Connection Flow

Practical: Connect and use dummy APIs with backend.

Chapter 7 — PHP Backend Development

- PHP Basics
- Server-side Scripting
- Form Handling
- GET & POST Methods
- Backend Logic

Practical: Build registration and login backend.

Chapter 8 — MySQL Database Integration

- MySQL Basics
- Tables & Relationships
- CRUD Operations
- Database Connectivity
- Connection Strings

Practical: Connect PHP with MySQL and perform CRUD operations.

Chapter 9 — Backend Security

- Password Hashing
- Prepared Statements
- Authentication Security
- Secure APIs
- SQL Injection Basics

Practical: Build secure backend authentication system.

Chapter 10 — Deployment & Final Project

- Backend Deployment
- Hosting Basics
- Cloud Database Connections
- Backend Project Structure

- Industry Workflow

Practical: Deploy complete backend project online.

Final Outcome:

Students will be able to:

- Build backend systems
- Create REST APIs
- Work with Firebase and MongoDB
- Build PHP + MySQL applications
- Handle JSON and APIs
- Create authentication systems
- Understand backend architecture
- Deploy backend applications